2023 年春季学期数学物理方程期末试题

姓名: 学号:

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本卷共 2 页, 含公式附录. 请将所有解答写在答题卷上.

1. (8 分) 以勒让德多项式为基，在区间 $[-1,1]$ 上将 $f(x)=x^3+1$ 展开为广义傅里叶级数.
2. (12) 对于 $y>0$ 和 $y<0$, 分别寻找合适的变量代换，将 Tricomi 方程化为标准形式:

$$y\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0.$$

3. (12 分) 求八分之一无界空间 $V=\{(x,y,z):x>0,y>0,z>0\}$ 内泊松方程的格林函数:

$$\begin{aligned}&\nabla^2 G(\mathbf{r};\mathbf{r}')=-\delta(\mathbf{r}-\mathbf{r}'),\quad \mathbf{r},\mathbf{r}'\in V,\\&G|_{\Sigma}=0;\end{aligned}$$

其中 $\mathbf{r}=(x,y,z)$, $\mathbf{r}'=(x',y',z')$, $\nabla^2=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$, Σ 代表 V 的边界.

4. (16 分) 用拉普拉斯变换法求解半无界弦的波动问题:

$$\begin{aligned}&\frac{\partial^2 u}{\partial t^2}-a^2\frac{\partial^2 u}{\partial x^2}=0,\qquad t>0,\,x>0,\\&u|_{x=0}=f(t),\quad t>0,\\&u|_{t=0}=\frac{\partial u}{\partial t}|_{t=0}=0.\end{aligned}$$

5. (16 分) 用傅里叶变换法求解上半平面问题:

$$\begin{aligned}&\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0,\qquad -\infty<x<\infty,\quad y>0,\\&u|_{y=0}=f(x),\quad u|_{y\to\infty}\text{有界}.\end{aligned}$$

6. (18 分) 求解球内问题:

$$\begin{aligned}&\nabla^2 u=r\sin\theta\cos\phi,\quad r<a,\\&u|_{r=a}=0,\quad u|_{r=0}<\infty.\end{aligned}$$

7. (18 分) 半径为 a、高为 h 的均匀圆柱体，上下底温度保持为 0, 侧面温度为常量 u_0, 求圆柱体内的稳定温度分布.

公式附录:

A. 考虑含勒让德多项式的积分 $I=\int_{-1}^{1}x^kP_l(x)dx$, 其中 k,l 为自然数。当 $k\pm l$ 为奇数时, $I=0$; $k<l$ 时, $I=0$; $k=l+2n$ $(n\geq 0)$ 时,

$$I=2^{l+1}\frac{(l+2n)!(l+n)!}{n!(2l+2n+1)!}.$$

B. 连带勒让德函数:

$$P_l^m(x)=\frac{1}{2^l l!}(-1)^m(1-x^2)^{m/2}[(x^2-1)^l]^{(l+m)}.$$
$$P_0^0(x)=1,\ P_1^0(x)=x,\ P_1^1(x)=-(1-x^2)^{1/2},$$
$$P_2^0(x)=\frac{1}{2}(3x^2-1),\ P_2^1(x)=-3x(1-x^2)^{1/2},\ P_2^2(x)=3(1-x^2).$$

C. 拉普拉斯变换:

$$\sin(\omega t)\leftrightarrow\frac{\omega}{p^2+\omega^2},\quad \cos(\omega t)\leftrightarrow\frac{p}{p^2+\omega^2},\quad \frac{1}{\sqrt{\pi t}}\leftrightarrow\frac{1}{\sqrt{p}},$$
$$\frac{1}{\sqrt{\pi t}}\exp(-\frac{a^2}{4t})\leftrightarrow\frac{e^{-a\sqrt{p}}}{\sqrt{p}},\quad \frac{a}{2\sqrt{\pi t^3}}\exp(-\frac{a^2}{4t})\leftrightarrow e^{-a\sqrt{p}}\quad (a>0).$$
$$H(t-a)\leftrightarrow\frac{e^{-ap}}{p},\quad e^{at}f(t)\leftrightarrow F(p-a),\quad f(t-a)H(t-a)\leftrightarrow e^{-ap}F(p).$$

其中 H 是单位阶跃函数, 即 $H(x\geq 0)=1, H(x<0)=0$. $F(p)$ 表示 $f(t)$ 的拉普拉斯变换.

D. 球贝塞尔函数:

$$j_0(x)=\frac{\sin x}{x},\quad n_0(x)=-\frac{\cos x}{x},\quad j_1(x)=\frac{\sin x-x\cos x}{x^2},\quad n_1(x)=-\frac{\cos x+x\sin x}{x^2}.$$

E. 变量代换:

$$a\frac{\partial^2u}{\partial x^2}+2b\frac{\partial^2u}{\partial x\partial y}+c\frac{\partial^2u}{\partial y^2}+d\frac{\partial u}{\partial x}+e\frac{\partial u}{\partial y}+fu+g=0$$

$$\leftrightarrow A\frac{\partial^2u}{\partial\xi^2}+2B\frac{\partial^2u}{\partial\xi\partial\eta}+C\frac{\partial^2u}{\partial\eta^2}+D\frac{\partial u}{\partial\xi}+E\frac{\partial u}{\partial\eta}+Fu+G=0,$$

$$\begin{aligned}
A &= a(\frac{\partial\xi}{\partial x})^2+2b\frac{\partial\xi}{\partial x}\frac{\partial\xi}{\partial y}+c(\frac{\partial\xi}{\partial y})^2,\\
B &= a\frac{\partial\xi}{\partial x}\frac{\partial\eta}{\partial x}+b(\frac{\partial\xi}{\partial x}\frac{\partial\eta}{\partial y}+\frac{\partial\xi}{\partial y}\frac{\partial\eta}{\partial x})+c\frac{\partial\xi}{\partial y}\frac{\partial\eta}{\partial y},\\
C &= a(\frac{\partial\eta}{\partial x})^2+2b\frac{\partial\eta}{\partial x}\frac{\partial\eta}{\partial y}+c(\frac{\partial\eta}{\partial y})^2,\\
D &= a\frac{\partial^2\xi}{\partial x^2}+2b\frac{\partial^2\xi}{\partial x\partial y}+c\frac{\partial^2\xi}{\partial y^2}+d\frac{\partial\xi}{\partial x}+e\frac{\partial\xi}{\partial y},\\
E &= a\frac{\partial^2\eta}{\partial x^2}+2b\frac{\partial^2\eta}{\partial x\partial y}+c\frac{\partial^2\eta}{\partial y^2}+d\frac{\partial\eta}{\partial x}+e\frac{\partial\eta}{\partial y},\\
F &= f,\quad G=g.
\end{aligned}$$